

(a)



(b)

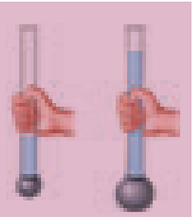


(c)

Figure 3.3 Pressure due to fluids

Similarly, liquid pressure acts in lateral sides also. When a bottle having water is pierced on the sides we can see water coming out with a speed as in Figure 3.3(c). This is because liquid exerts lateral pressure on the walls the container.

Activity 2



Take a transparent plastic pipe. Also take a balloon and tie it tightly over one end of the plastic pipe. Pour some water in the pipe from the top. What happens? The balloon tied at the bottom stretches and bulges out. It shows that the water poured in the pipe exerts a pressure on the bottom of its container.

3.2.2 Factors determining liquid pressure in liquids

Pressure exerted by a liquid at a point is determined by,

(i) depth (h)

(ii) density of the liquid (ρ)

(iii) acceleration due to gravity (g).

Activity 3



Take a large plastic can. Punch holes with a nail in a vertical line on the side of the can as shown in figure. Then fill the can with water. The water may just dribble out from the top hole, but with increased speed at the bottom holes as depth causes the water to squirt out with more pressure.

From this activity we can infer that pressure varies as depth increases. But, it is same at a particular depth independent of the direction.

Activity 4

Take two liquids of different densities say water and oil to a same level in two plastic containers. Make holes in the two containers at the same level. What do you see? It can be seen that water is squirting out with more pressure than oil. This indicates that pressure depends on density of the liquid.



3.2.3 Pressure due to a liquid column

A tall beaker is filled with liquid so that it forms a liquid column. The area of cross section at the bottom is A. The density of the liquid is ρ. The height of the liquid column is h. In other words the depth of the water from the top level surface is 'h' as shown in Figure 3.4.

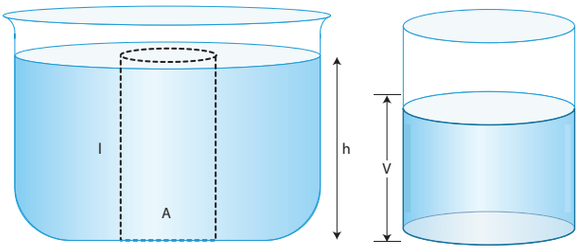


Figure 3.4 Pressure due to a liquid column

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